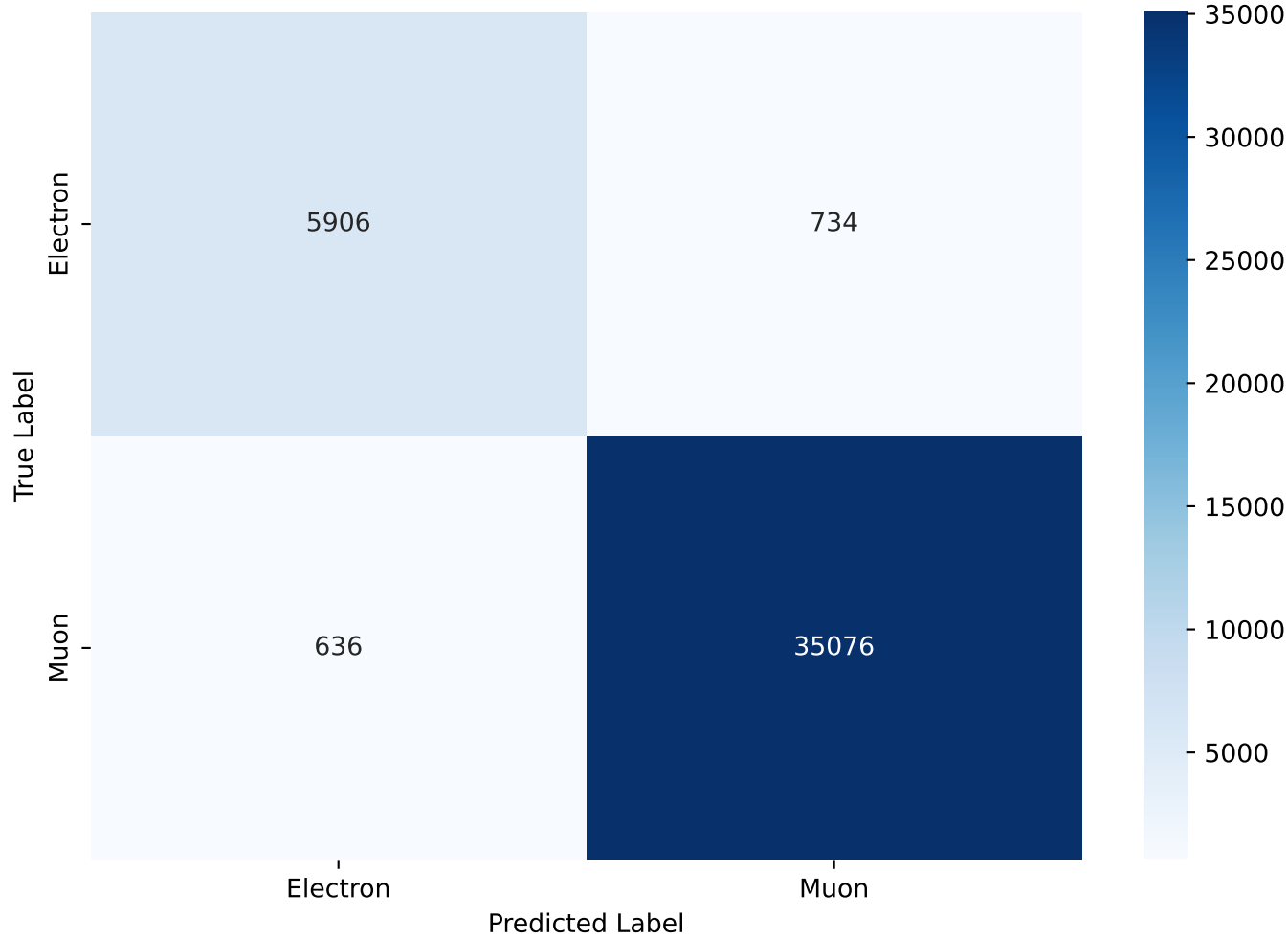
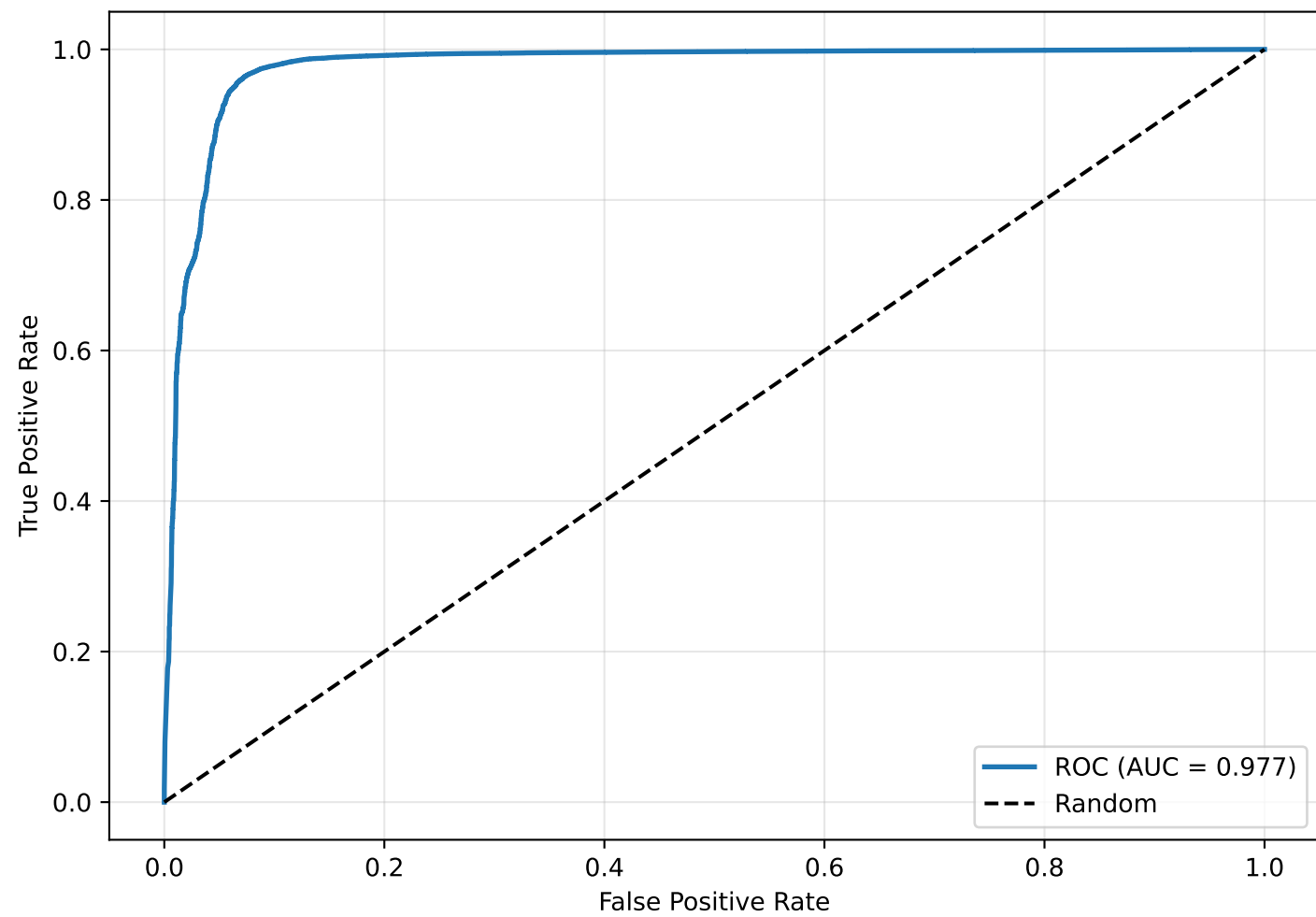


**EtaRange = FullRange - Accuracy on MC: 0.968
(trained on full TrueData)**

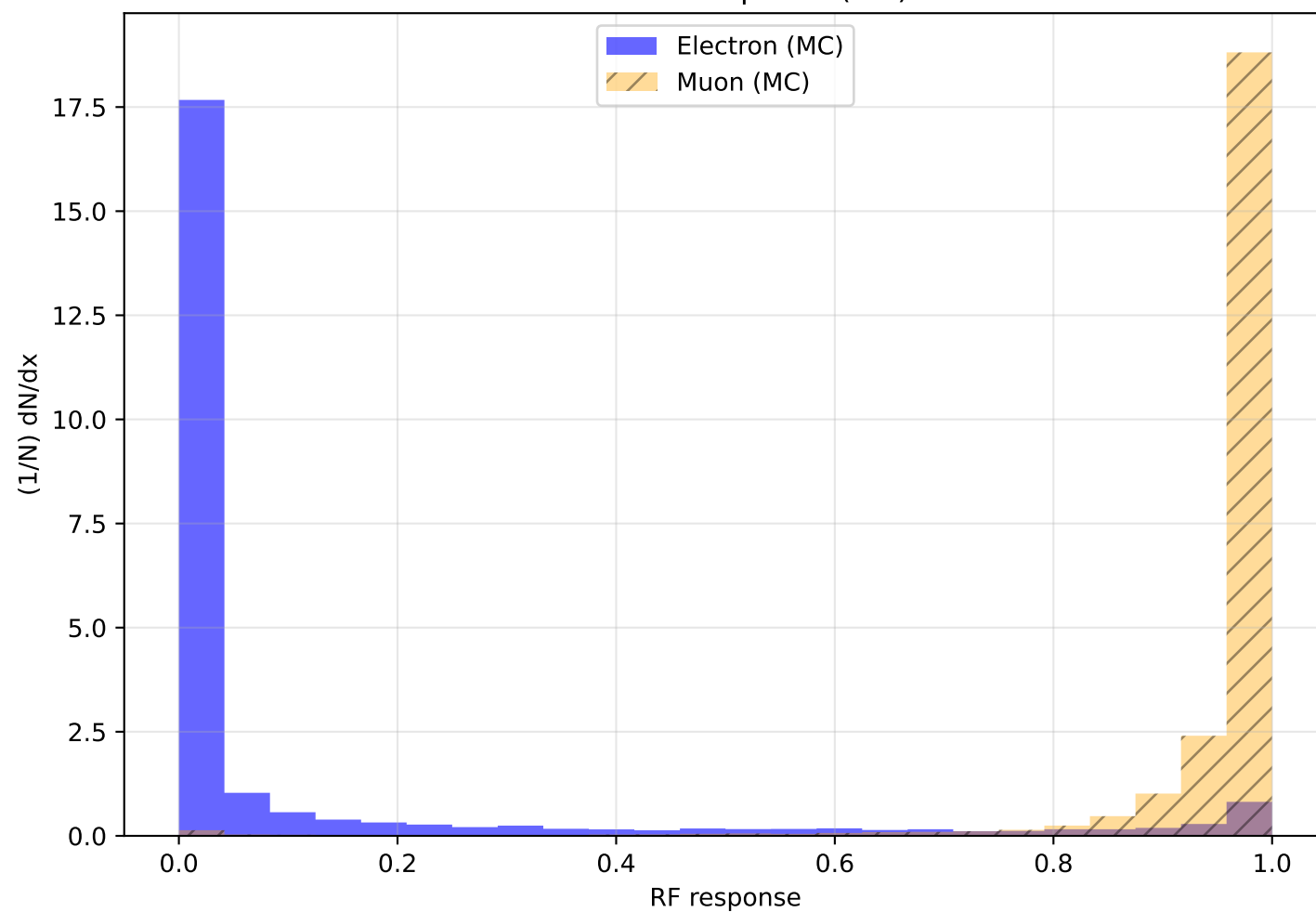
Confusion Matrix



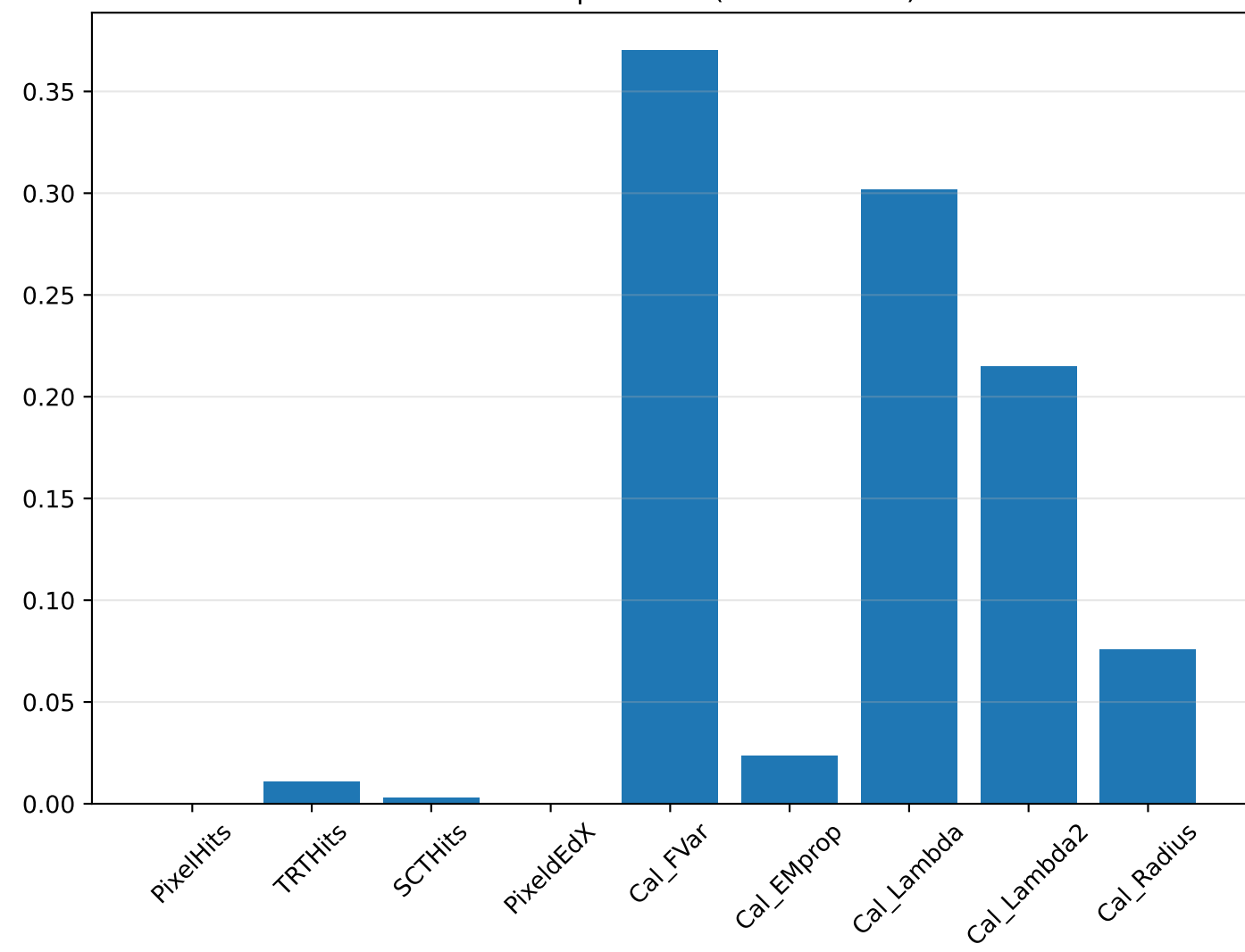
ROC Curve



Classifier Response (MC)

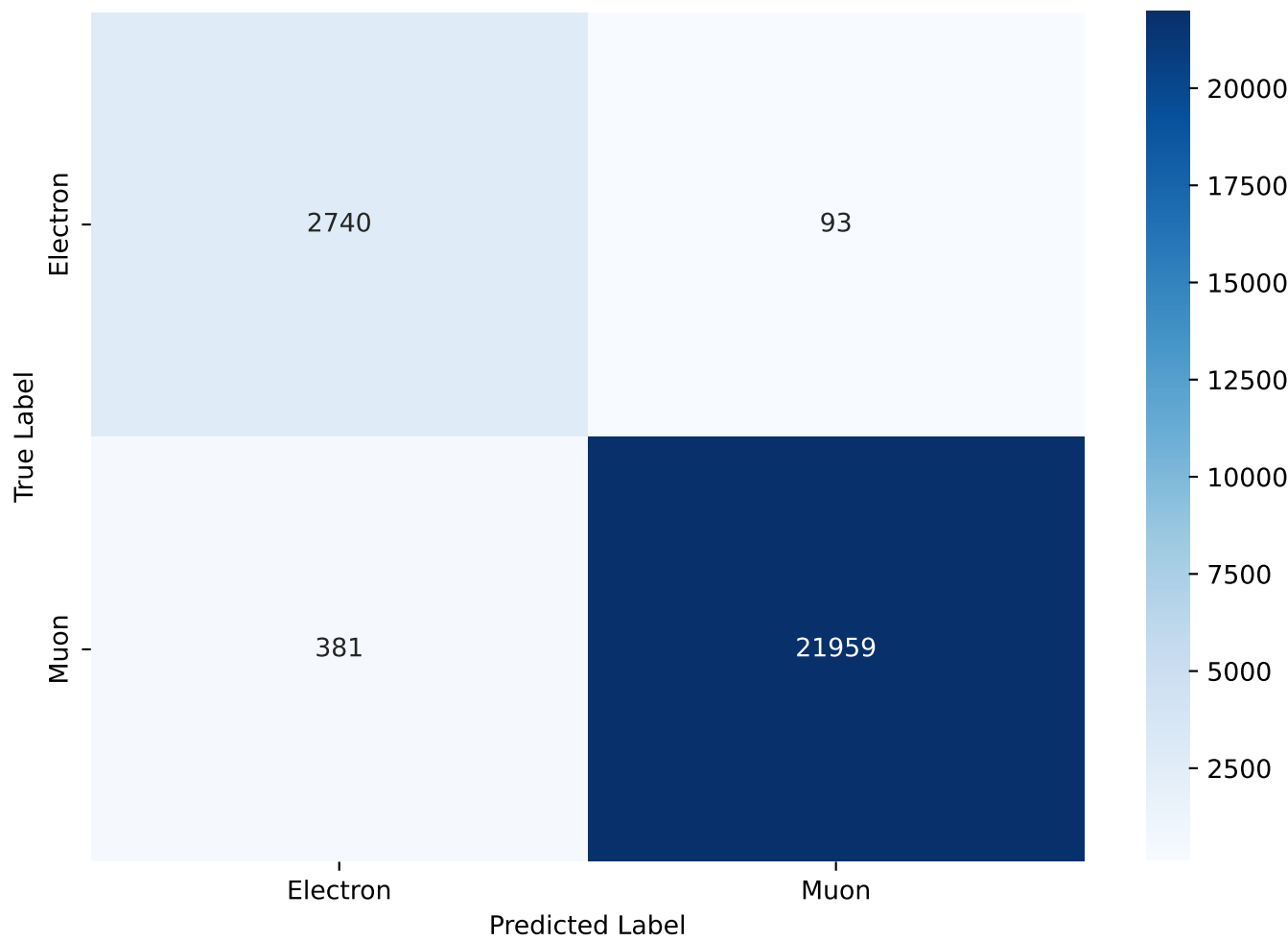


Feature Importance (full TrueData)

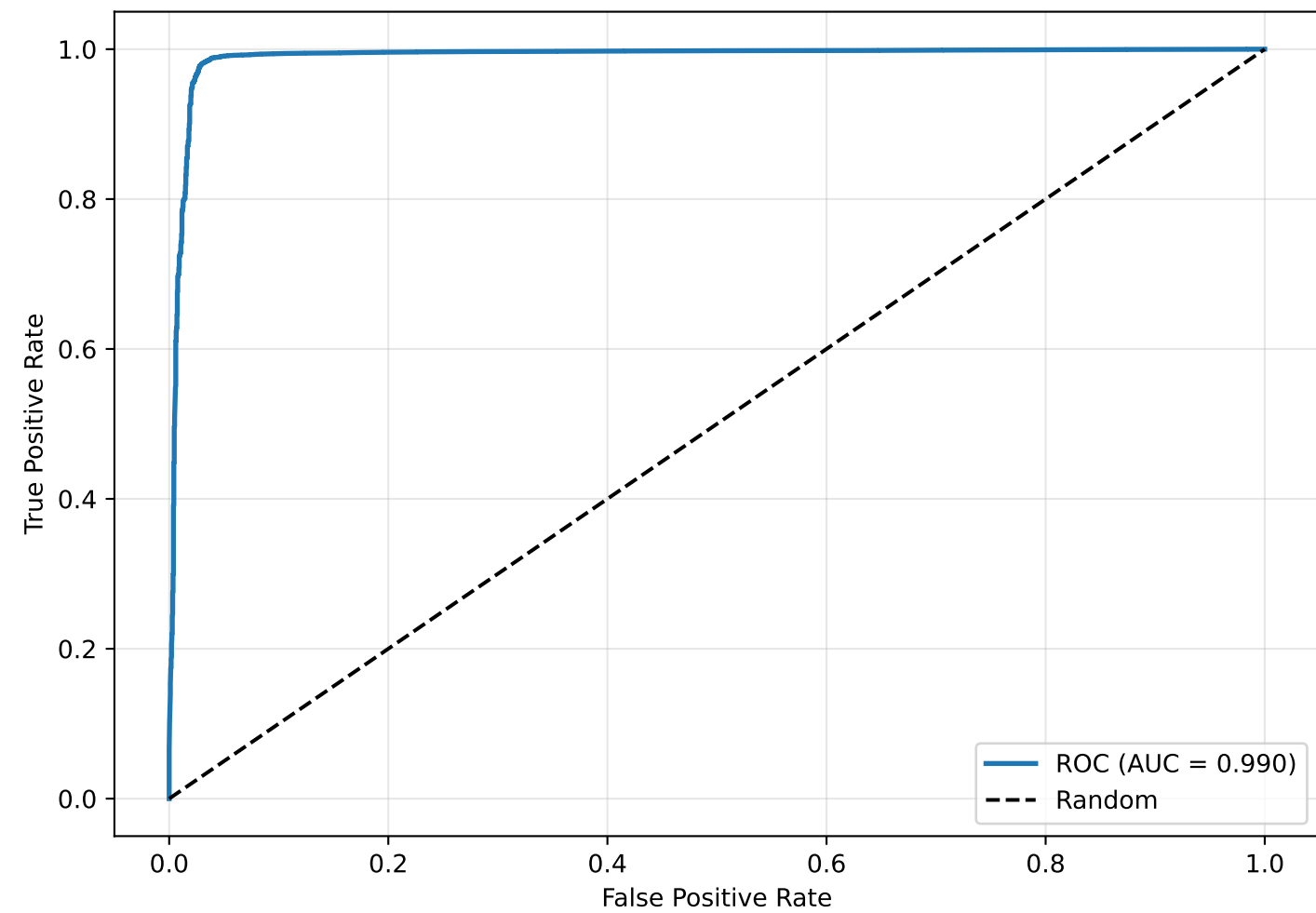


EtaRange = <1.0 - Accuracy on MC: 0.981 (trained on full TrueData)

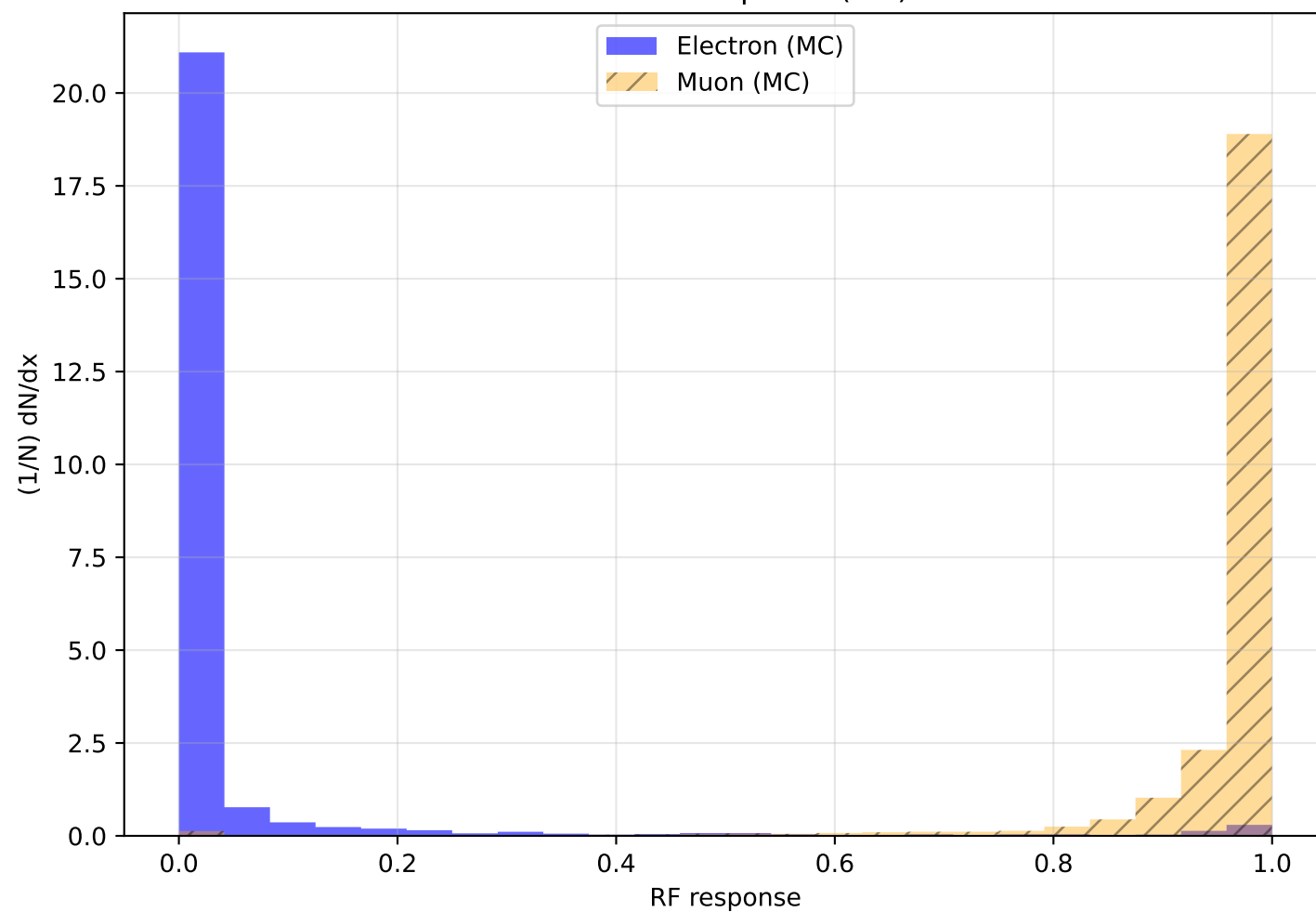
Confusion Matrix



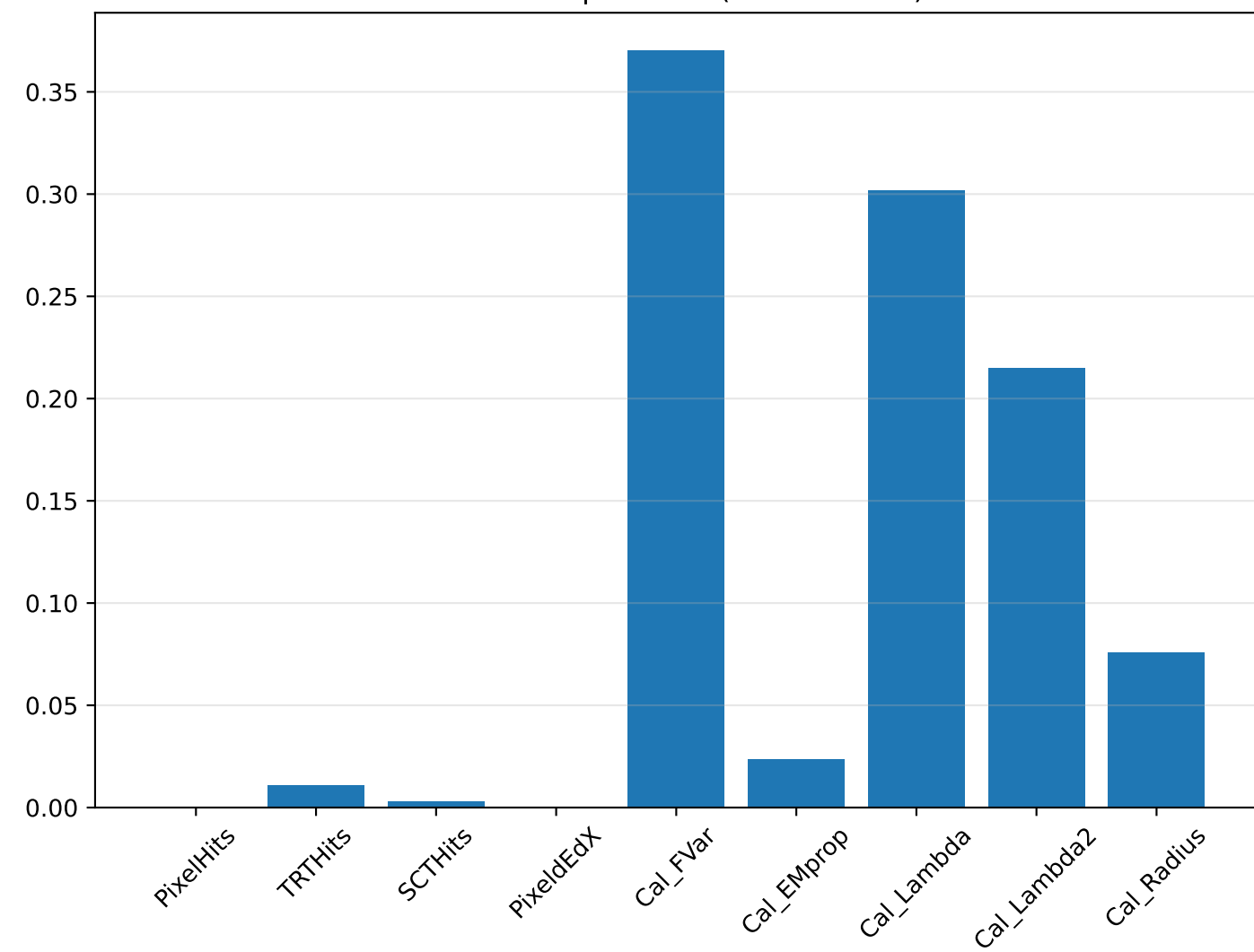
ROC Curve



Classifier Response (MC)

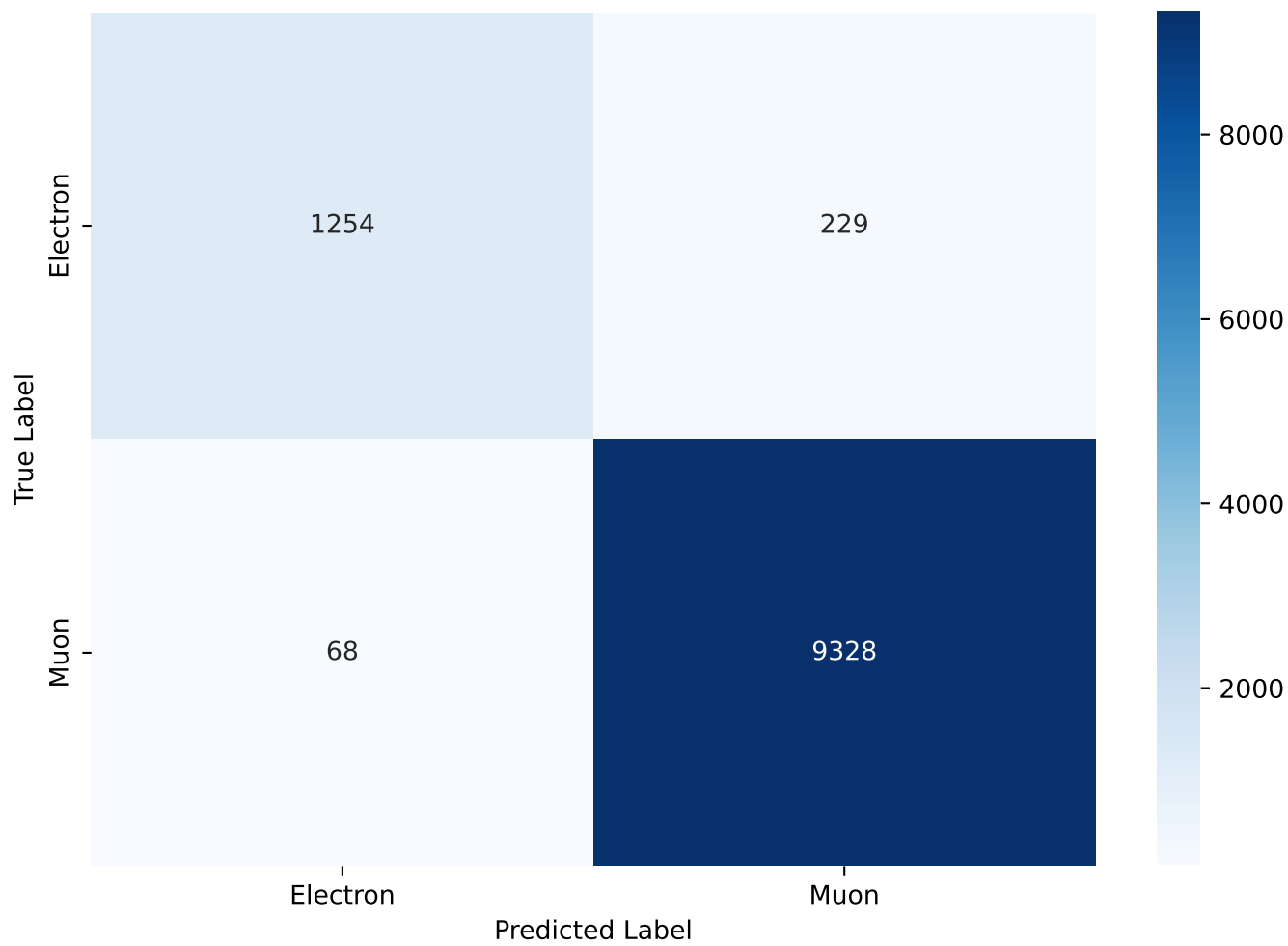


Feature Importance (full TrueData)

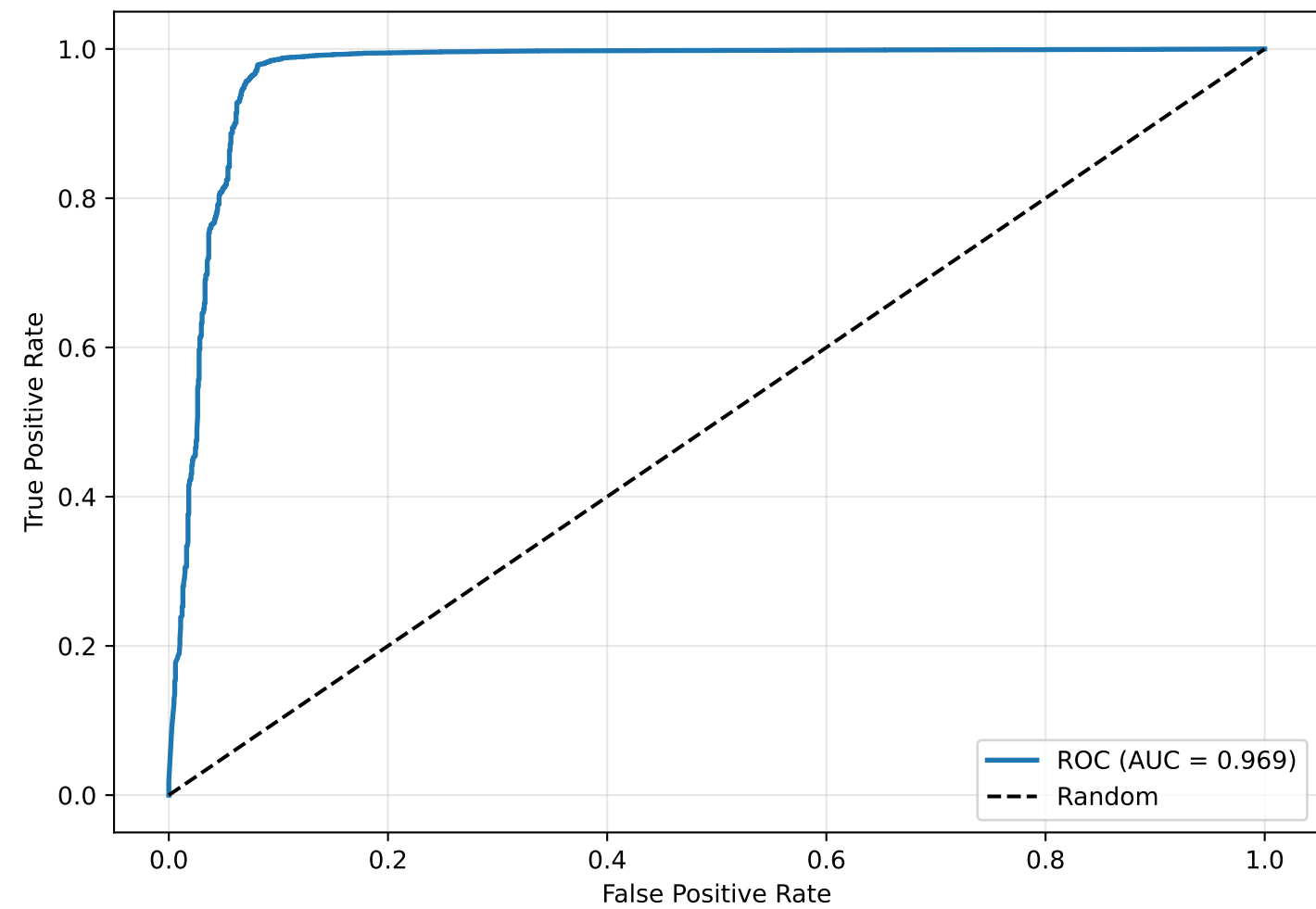


EtaRange = 1.0-1.5 - Accuracy on MC: 0.973
(trained on full TrueData)

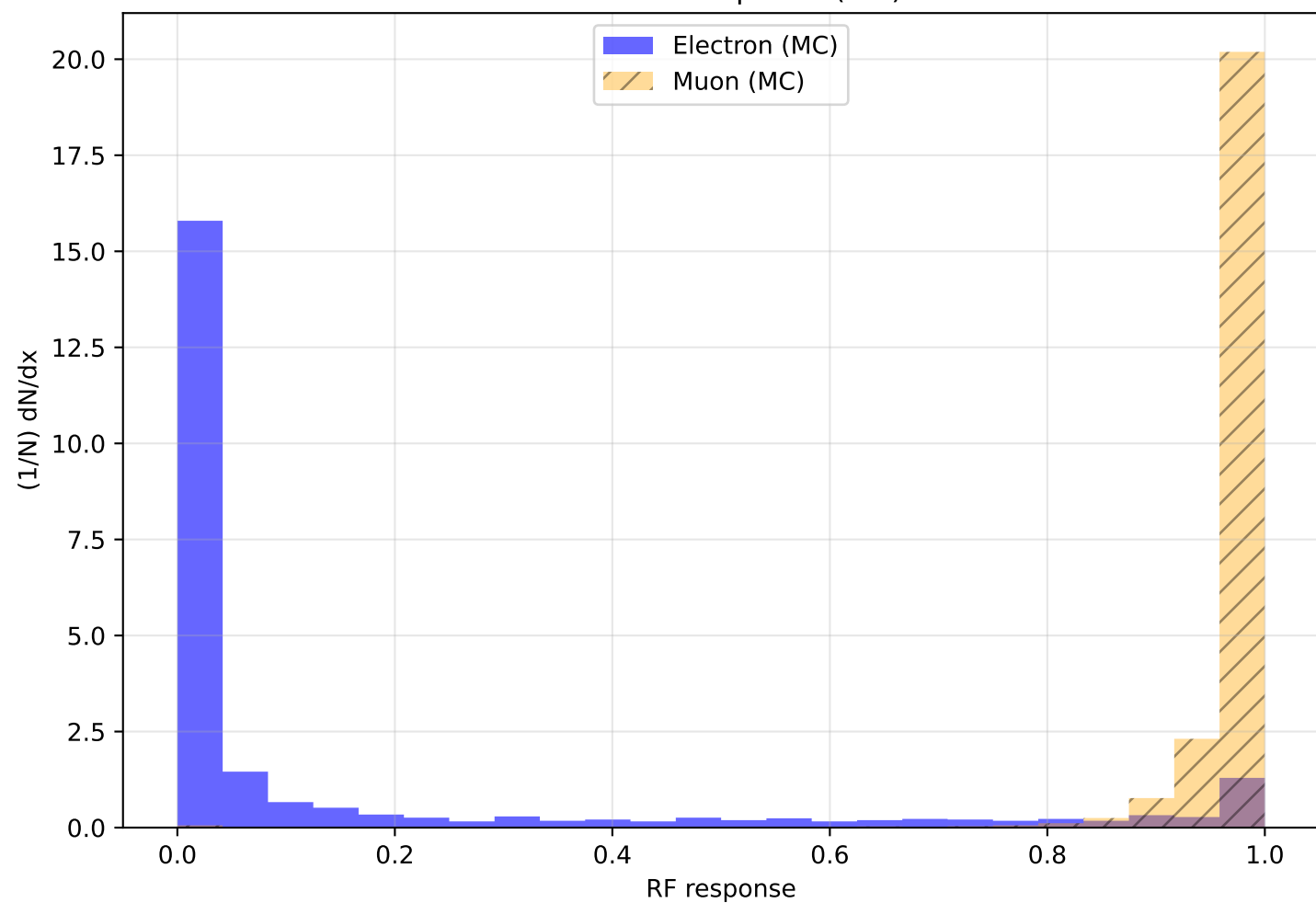
Confusion Matrix



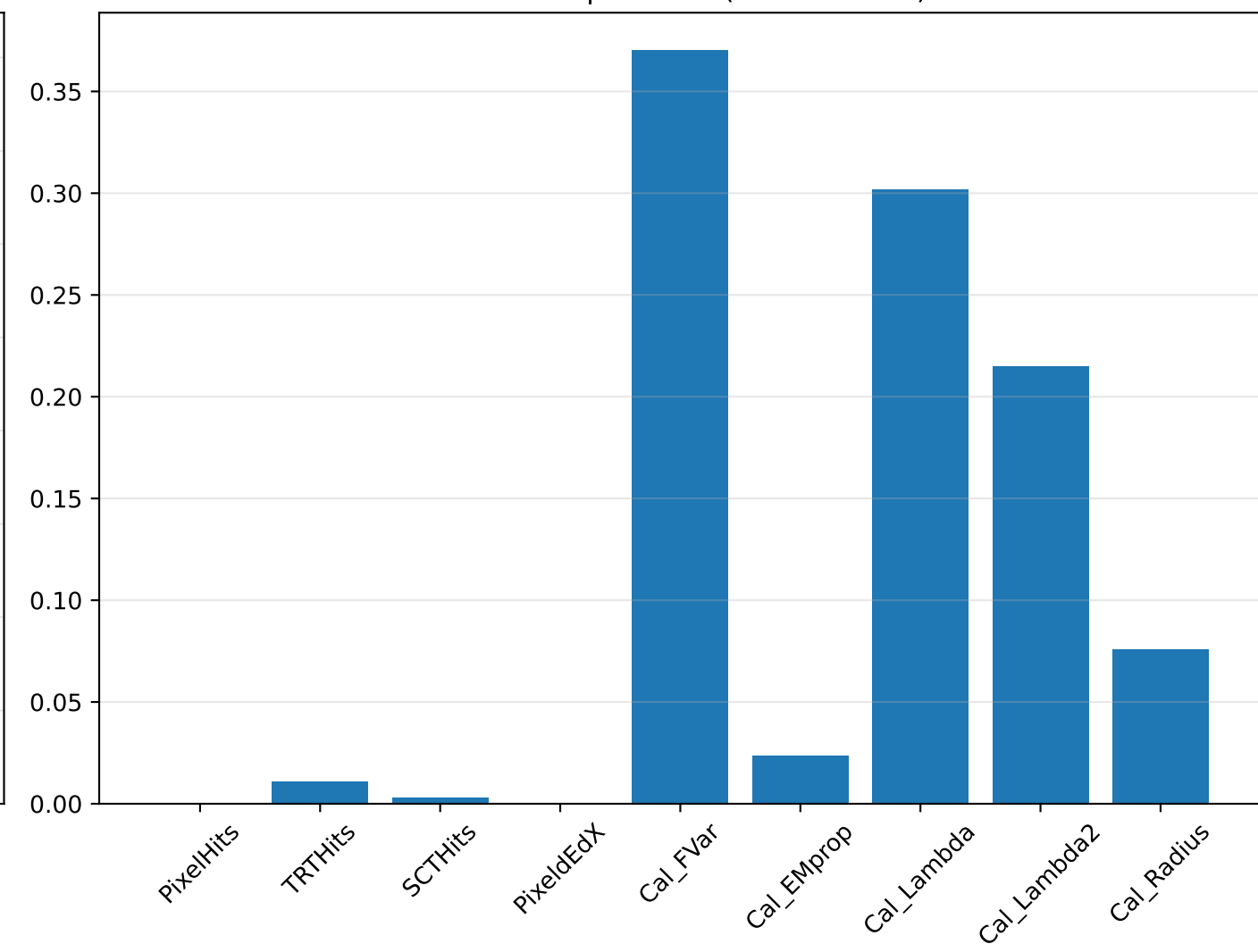
ROC Curve



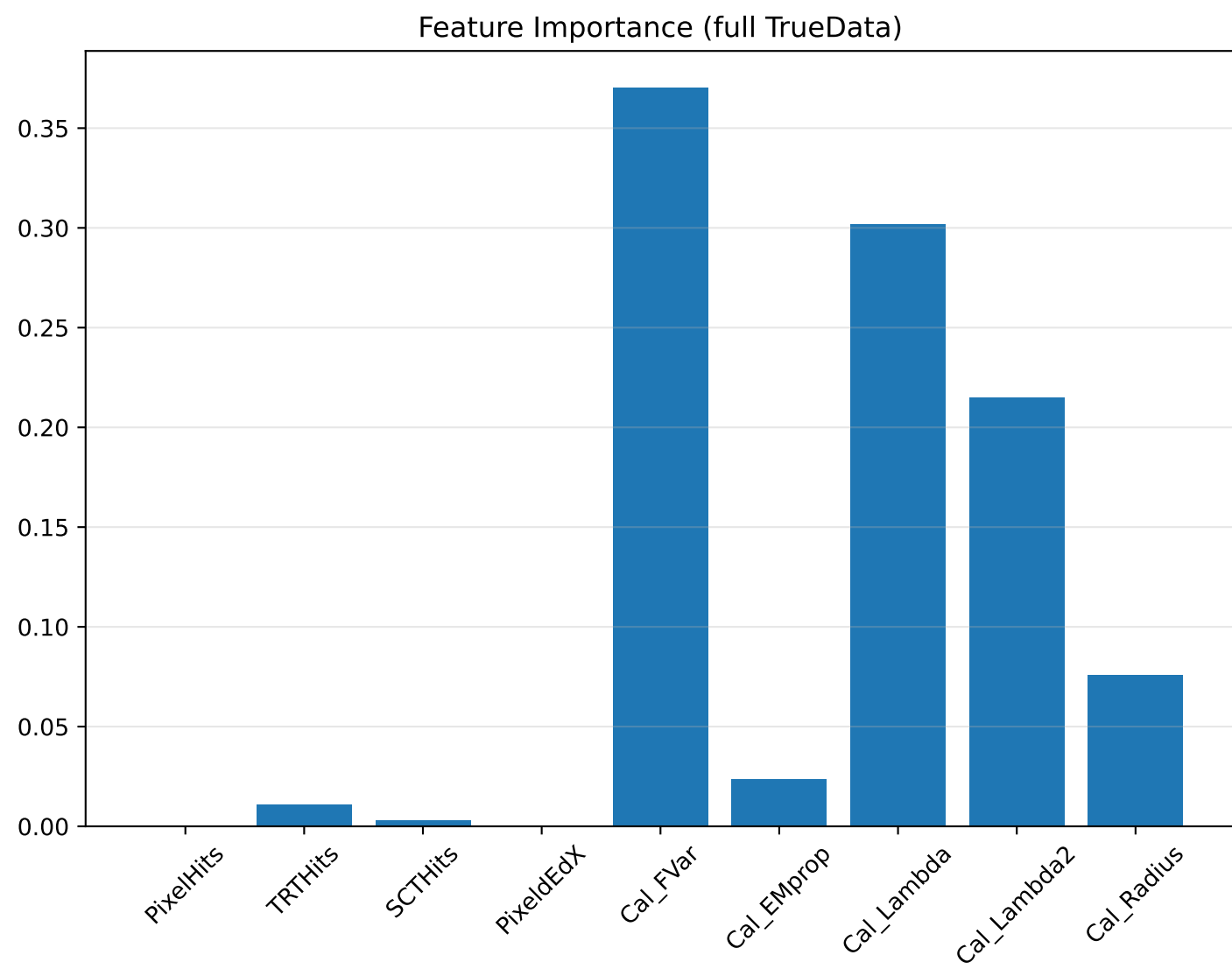
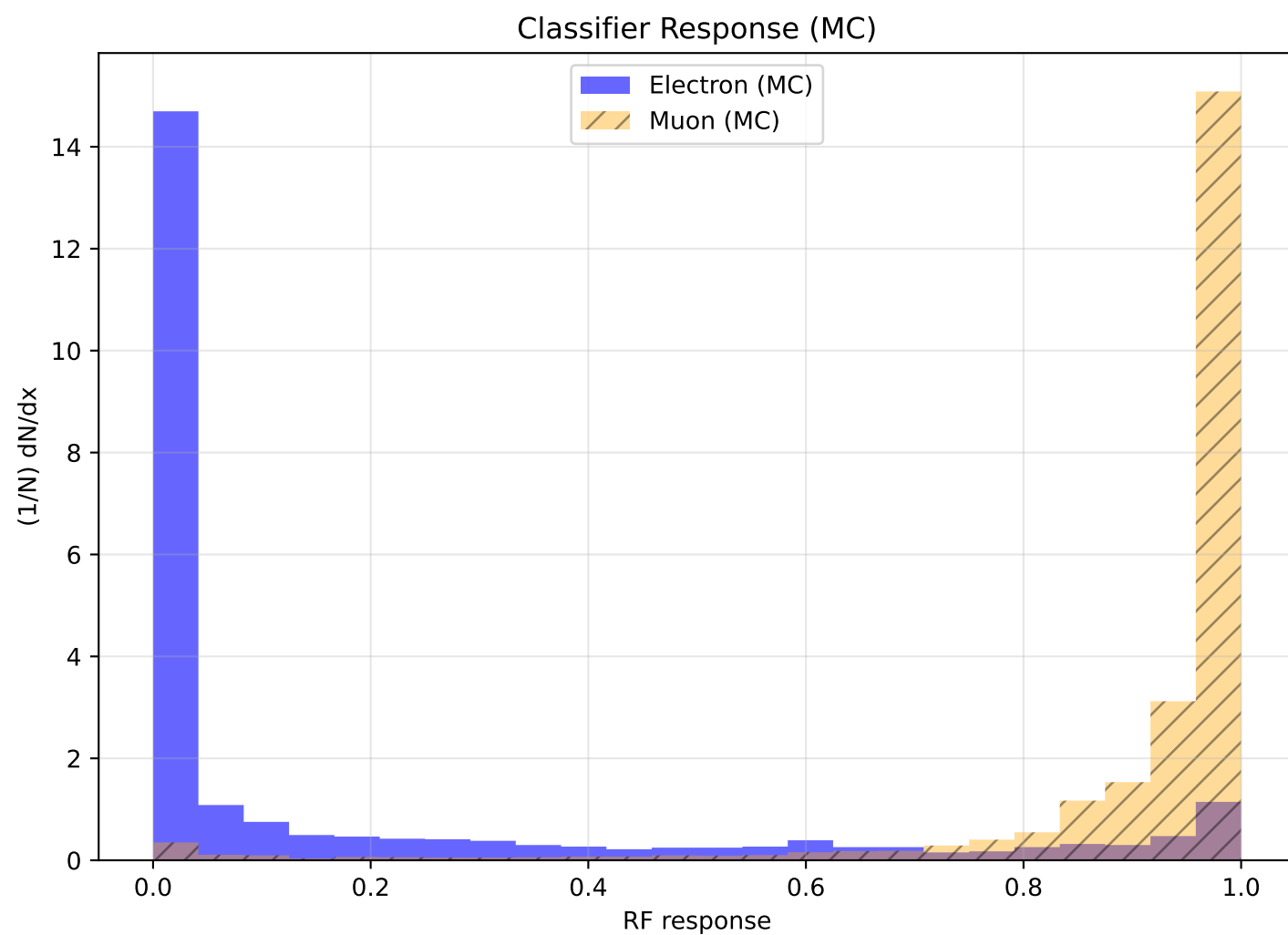
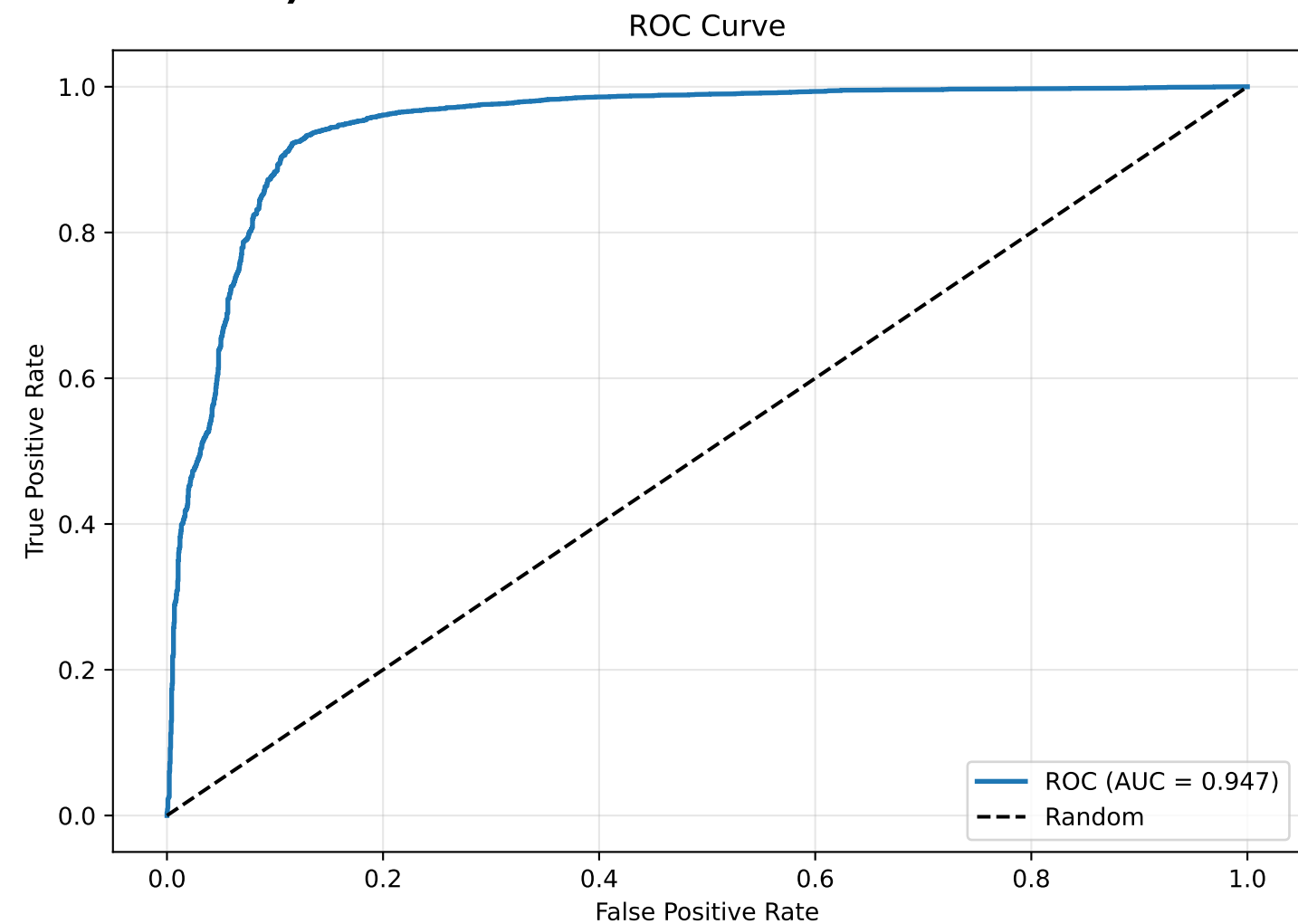
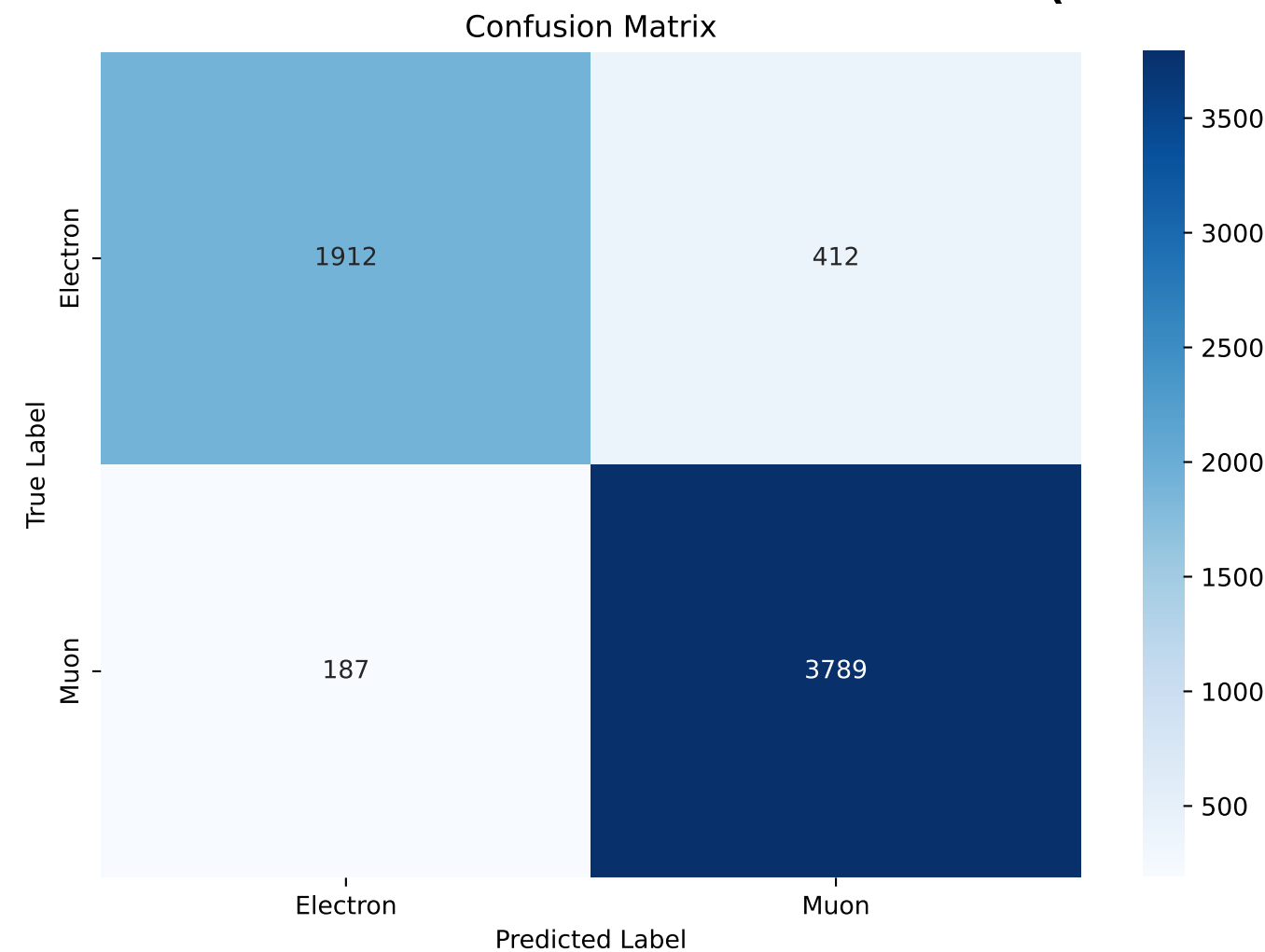
Classifier Response (MC)



Feature Importance (full TrueData)



EtaRange = 1.5-2.5 - Accuracy on MC: 0.905 (trained on full TrueData)



Classification Reports Summary

Trained: Double TrueData.root | Test: MC Samples

EtaRange = FullRange

	precision	recall	f1-score	support
Electron	0.90	0.89	0.90	6640
Muon	0.98	0.98	0.98	35712
accuracy			0.97	42352
macro avg	0.94	0.94	0.94	42352
weighted avg	0.97	0.97	0.97	42352

EtaRange = <1.0

	precision	recall	f1-score	support
Electron	0.88	0.97	0.92	2833
Muon	1.00	0.98	0.99	22340
accuracy			0.98	25173
macro avg	0.94	0.98	0.95	25173
weighted avg	0.98	0.98	0.98	25173

EtaRange = 1.0-1.5

	precision	recall	f1-score	support
Electron	0.95	0.85	0.89	1483
Muon	0.98	0.99	0.98	9396
accuracy			0.97	10879
macro avg	0.96	0.92	0.94	10879
weighted avg	0.97	0.97	0.97	10879

EtaRange = 1.5-2.5

	precision	recall	f1-score	support
Electron	0.91	0.82	0.86	2324
Muon	0.90	0.95	0.93	3976
accuracy			0.90	6300
macro avg	0.91	0.89	0.90	6300
weighted avg	0.91	0.90	0.90	6300